DEPARTMENT OF APPLIED SCIENCES

Discrete Structure & Theory of Logics

UNIT – III

Lattices and Boolean Algebras

What do we study in Discrete Structure & Theory of Logics?

Discrete Structure & Theory of Logics is a branch of mathematics and computer science that deals with discrete (countable) objects and logical reasoning. It provides the mathematical foundation for computer science, programming, algorithms, artificial intelligence, databases, and digital circuit design.

Why do we need to study Discrete Structure & Theory of Logics?

We study Discrete Structure & Theory of Logics to develop logical thinking, mathematical reasoning, and problem-solving skills**.** It provides the foundation for programming, algorithms, data structures, databases, artificial intelligence, cryptography, and digital circuits**,** making it an essential subject in computer science.

Where do we use Discrete Structure & Theory of Logics?

Discrete Structure & Theory of Logics is used in computer programming, algorithm design, data structures, databases, artificial intelligence, computer networks, cybersecurity, cryptography, digital circuit design, software engineering, and web technologies. It helps in developing efficient, secure, and reliable computing systems.

CONTENTS

Lattices and Boolean Algebras: Posets, Hasse diagram, Extremal elements; Properties of Lattices; Bounded, Complemented, Distributed, and Modular Lattices; Boolean Algebras;Important Examples; Some other properties.

3.0 poset : Suppose R is a relation on a set S satisfying the following three properties:

[1] (Reflexive) For any $a \in S$, we have aRa.
[2] (Antisymmetric) If aRb and bRa, then $a = b$.
[3] (Transitive) If aRb and bRc, then aRc.

Then R is called a partial order or, simply an order relation, and R is said to define a partial ordering of S. The set S with the partial order is called a partially ordered set or, simply, an ordered set or poset.

Remark : A partial order relation is usually denoted by $\preccurlyeq$; and $a \preccurlyeq b$ is read "a precedes b." In this case we also write:

$a \prec b$ means $a \preccurlyeq b$ and $a \neq b$; read "a strictly precedes b."

b ≽ a means a ≼ b; read "b succeeds a."

b ≻ a means a ≺ b; read "b strictly succeeds a."

When there is no ambiguity, the symbols ≤ , < , > and ≥ are frequently used instead of ≼,≺ , ≻, and ≽ , respectively.

Comparability, Linearly Ordered Sets:

Suppose a and b are elements in a partially ordered set S. We say a and b are comparable if a ≼ b or b ≽ a that is, if one of them precedes the other . Thus a and b are noncomparable , written a ∥ b if neither a ≼ b nor b ≽ a .

The word "partial" is used in defining a partially ordered set S since some of the elements of S need not be comparable. Suppose, on the other hand, that every pair of elements of S are comparable. Then S is said to be totally ordered or linearly ordered, and S is called a chain. Although an ordered set S may not be linearly ordered, it is still possible for a subset A of S to be linearly ordered. Clearly, every subset of a linearly ordered set S must also be linearly ordered.

Example1 : Verify that the Divisibility Relation is a Partial Order

Let $S=\{1,2,3,6\}$. Define a relation R on S by $aRb \Leftrightarrow$ a divides b, that is, $aRb \Leftrightarrow a/b$.

Show that R is a partial order on S.

Solution: A relation is a partial order if it is reflexive, antisymmetric, and transitive**.**

Reflexive Every element divides itself $1/1$, $2/2$, $3/3$, $6/6$. Hence, $aRa, \forall a \in S$. Therefore, R is reflexive.

Antisymmetric Property A relation R is antisymmetric if aRb and $bRa \Longrightarrow a=b, \forall a,b \in S$.

Suppose aRb and bRa. Since the relation R is defined by divisibility, a/b and b/a. Therefore, there exist integers m and n such that $b=am$ and $a=bn$. Substituting $b=am$ into $a=bn$, we get $a=(am)n$

Since $a \neq 0$, $1=mn$. The only positive integers satisfying $mn=1$ are $m=1,n=1$. Hence, $a=b$. Therefore, R is antisymmetric.

Transitive A relation R is transitive if aRb and $bRc \Longrightarrow aRc, \forall a,b,c \in S$.

Suppose aRb and bRc . Since the relation R is defined by divisibility, a/b and b/c .Then there exist integers m and n such that $b=am$ and $c=bn$. Substituting $b=am$

into $c=bn$, we get $c=(am)n$. Since mn is an integer, $a|c$. Therefore, aRc. Hence,R is transitive.

Since R satisfies Reflexive , Antisymmetric and Transitive properties so the relation R is a partial order relation**.**

Hence, (S,R) is a Partially Ordered Set (Poset).

Example 2: Show that the Subset Relation is a Partial Order
Let $S = P(\{1,2\}) = \{\emptyset, \{1\}, \{2\}, \{1,2\}\}$, where $P(\{1,2\})$ denotes the power set of {1,2}. Define a relation R on S by $ARB \iff A \subseteq B$. Show that R is a partial order on S .

Solution To prove that R is a partial order, we verify the three properties.

Reflexive Property A relation R is reflexive if $\boldsymbol{ARA}, \forall \boldsymbol{A} \in \boldsymbol{S}$. Suppose $A \in S$. Every set is a subset of itself. Therefore, $\boldsymbol{A} \subseteq \boldsymbol{A}$. Hence, , $\forall \boldsymbol{A} \in \boldsymbol{S}$. Thus, R is reflexive.
Antisymmetric Property
$\boldsymbol{A\ relation\ R\ is}$ *antisymmetric* $\boldsymbol{if\ ARB\ and\ BRA \implies A = B, \forall A, B \in S}$. Suppose $\boldsymbol{ARB}$ and BRA. Then $A \subseteq B$ and $B \subseteq A$. If each set is a subset of the other, then both sets contain exactly the same elements. Hence, A=B. Therefore, *R* is antisymmetric.

Transitive PropertyA relation R is transitive if $\boldsymbol{ARB}$ and $\boldsymbol{BRC} \implies \boldsymbol{ARC}$, $\forall A,B,C \in S$. Suppose $\boldsymbol{ARB}$ and $\boldsymbol{BRC}$. Then $A \subseteq B$ and $B \subseteq C$. Since every element of A belongs to B, and every element of B belongs to C, every element of A also belongs to C. Therefore, $A \subseteq C$. Hence, $\boldsymbol{ARC}$. Thus, *R* is transitive.

Since the relation *R* satisfies reflexive , antisymmetric and transitive properties therefore, R is a partial order relation. Hence,
(S,R) is a Partially Ordered Set (Poset).

Example 3: Let R be the relation of divisibility on the set of integers Z , defined by $R = \{(a,b) \in Z \times Z : a \mid b\}$. Determine whether (Z, R) is a partially ordered set (Poset**)**.

Solution: The relation " | " of divisibility is not an ordering of the set Z of integers. Specifically, the relation is not antisymmetric. For instance, $\mathbf{2} \mid -\mathbf{2}$ and $-\mathbf{2} \mid \mathbf{2}$, but $\mathbf{2} = -\mathbf{2}$.

Example 4 : Let A={1,2,3} . Define the relation R={(1,1),(2,2),(3,3),(1,2),(1,3)}. Show that (A,R) is a Partially Ordered Set (POSET).

Solution: To prove that (A,R) is a POSET, we check the following three properties**.**
A relation R on a set A is **reflexive** if every element is related to itself i.e. $\forall\ \boldsymbol{a} \in \boldsymbol{A},\ (\boldsymbol{a}, \boldsymbol{a}) \in \boldsymbol{R}$. Since (1,1), (2,2), (3,3) ∈ R every element is related to itself. Hence, R is reflexive.

A relation R is **antisymmetric** if $(a,b) \in R \text{ and } (b,a) \in R \implies a = b$.

Here, $(1,2) \in R$, but $(2,1) \notin R$. $(1,3) \in R$ but $(3,1) \notin R$. There is no pair of distinct elements related in both directions. Hence, R is antisymmetric.

A relation R is **transitive** if $(a,b) \in R,\ (b,c) \in R \implies (a,c) \in R$. There is no pair such as (1,2) and (2,3) that requires another ordered pair. Therefore, the transitive condition is satisfied automatically. Hence, R is transitive.

Since the relation is reflexive , antisymmetric and Transitive .Therefore,(A,R) is POSET.

Practice Questions:

Question 1: Let $\boldsymbol{A} = \{\mathbf{1},\mathbf{2},\mathbf{4},\mathbf{8}\}$. Define the relation R on A by $\boldsymbol{aRb} \iff \boldsymbol{a} \mid \boldsymbol{b}$ (where a divides b). Show whether $(\boldsymbol{A},\boldsymbol{R})$ is a Partially Ordered Set (POSET) .

Question 2: Let $\boldsymbol{A} = \{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\}$. Define the relation $\boldsymbol{R} = \{(\boldsymbol{a},\boldsymbol{a}),(\boldsymbol{b},\boldsymbol{b}),(\boldsymbol{c},\boldsymbol{c}),(\boldsymbol{a},\boldsymbol{b}),(\boldsymbol{b},\boldsymbol{c}),(\boldsymbol{a},\boldsymbol{c})\}$. Determine whether $(\boldsymbol{A},\boldsymbol{R})$ is a POSET.

Question 3: Let $\boldsymbol{A} = \{\mathbf{1},\mathbf{2},\mathbf{3}\}$ Define the relation $\boldsymbol{R} = \{(\mathbf{1},\mathbf{1}),(\mathbf{2},\mathbf{2}),(\mathbf{3},\mathbf{3}),(\mathbf{1},\mathbf{2}),(\mathbf{2},\mathbf{1})\}$. Determine whether $(\boldsymbol{A},\boldsymbol{R})$ is a POSET.

Question 4: Let $\boldsymbol{A} = \{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\}$ Define the relation $\boldsymbol{R} = \{(\boldsymbol{a},\boldsymbol{a}),(\boldsymbol{b},\boldsymbol{b}),(\boldsymbol{c},\boldsymbol{c}),(\boldsymbol{a},\boldsymbol{b}),(\boldsymbol{b},\boldsymbol{c})\}$. Determine whether $(\boldsymbol{A},\boldsymbol{R})$ is a POSET.

3.1 Definition: Let $\boldsymbol{S}$ be a partially ordered set, and suppose $\boldsymbol{a},\boldsymbol{b}$ belong to $\boldsymbol{S}$. We say that a is an immediate predecessor of $\boldsymbol{b}$, or that $\boldsymbol{b}$ is an immediate successor of $\boldsymbol{a}$, or that $\boldsymbol{b}$ is a cover of a, written $\boldsymbol{a} \ll \boldsymbol{b}$ if $\boldsymbol{a} < \boldsymbol{b}$ but no element in $\boldsymbol{S}$ lies between $\boldsymbol{a}$ and $\boldsymbol{b}$, $\boldsymbol{i.e.}$ There exists no element c in S such that $\boldsymbol{a} < \boldsymbol{c} < \boldsymbol{b}$.

Suppose S is a finite partially ordered set. Then the order on S is completely known once we know all pairs $\boldsymbol{a},\boldsymbol{b}$ in S such that $\boldsymbol{a} \ll \boldsymbol{b}$, that is, once we know the relation $\ll$ on $\boldsymbol{S}$. This follows from the fact that x < y if and only if $\boldsymbol{x} \ll \boldsymbol{y}$ or there exist elements $\mathbf{a_1}$, $\mathbf{a_2}$,..., $\mathbf{a_m}$ in S such that x$\ll \mathbf{a_1} \ll \mathbf{a_2} \ll \cdots \ll \mathbf{a_m} \ll$ y .

3.2 HASSE DIAGRAMS OF PARTIALLY ORDERED SETS

The Hasse diagram of a finite partially ordered set $\boldsymbol{S}$ is the directed graph whose vertices are the elements of $\boldsymbol{S}$ and there is a directed edge from *a* to *b* whenever a $\ll$ b in $\boldsymbol{S}$. (Instead of drawing an arrow from *a* to *b*, we sometimes place *b* higher than *a* and draw a line between them. It is then understood that movement upwards indicates succession). In the diagram thus created, there is a directed edge from vertex *x* to vertex *y* if and only if *x*

$\ll$ *y*. Also, there can be no (directed) cycles in the diagram of *S* since the order relation is antisymmetric.

The Hasse diagram of a poset S is a picture of S; hence it is very useful in describing types of elements in *S*. Sometimes we define a partially ordered set by simply presenting its Hasse diagram. We note that the Hasse diagram of a poset *S* need not be connected**.**

Example 1: Draw the Hasse diagram for the poset (a,/),where a={1,3,9,27,81} ?

Solution: Since **1 << 3 3 << 9 9 << 27 27 <<**81 .So the Hasse diagram is



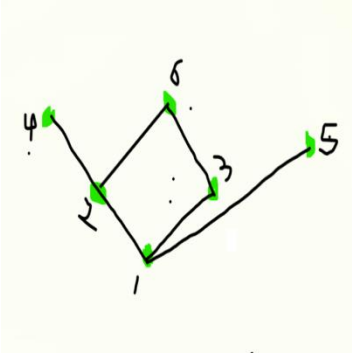
Example 2: – Let X ={1,2,3,4,5,6} and "/ " is a partial order relation on X .Draw the Hasse diagram of (X, /) .

Solution:

Since

- 1<<2, 1<<3, 1<<5
- 2<<4, 2<<6
- 3 << 6

So the Hasse diagram is

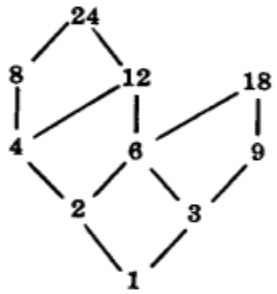


Example 3: Let $\boldsymbol{A} = \{\mathbf{1}, \mathbf{2}, \mathbf{3}, \mathbf{4}, \mathbf{6}, \mathbf{8}, \mathbf{9}, \mathbf{12}, \mathbf{18}, \mathbf{24}\}$ be ordered by the relation "*x* divides *y*." Draw the Hasse diagram .

Solution: Since

- 1<< 2 , 1<< 3
- 2<< 4, 2 << 6
- 3<<6, 3 << 9
- 4 << 8, 4 << 12
- 6 <<12, 6 << 18
- 8 << 24
- 9 << 18 , 12 << 24

So the Hasse diagram is



Example 4 : Consider the set $\boldsymbol{A} = \{\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}, \boldsymbol{u}, \boldsymbol{v}\}$ with the order relation $\boldsymbol{x} < \boldsymbol{y} < \boldsymbol{z} < \boldsymbol{u} <$

v. Draw the Hasse diagram .

Solution: Fig. shows the diagram of a chain with five elements



Practice Questions:

Question 1: Let $\boldsymbol{P} = \{\mathbf{1},\mathbf{2},\mathbf{3},\mathbf{4},\mathbf{5}\}$ and $\leq$ be the relation ‖less than or equal to‖ then draw the Hasse diagram.

Question 2: Let $\boldsymbol{X} = \{\mathbf{2},\mathbf{3},\mathbf{6},\mathbf{12},\mathbf{24},\mathbf{36}\}$, and the relation $\leq$ be such that $\boldsymbol{x} \leq \boldsymbol{y}$ if x divides y. Draw the Hasse diagram of $(\boldsymbol{X},\leq)$.

Question 3: Let $\boldsymbol{A}$ be a finite set and $\boldsymbol{P}(\boldsymbol{A})$ be its power set. Let $\subseteq$ be the inclusion relation on the elements of $\boldsymbol{P}(\boldsymbol{A})$. Draw the Hasse diagram of $(\boldsymbol{P}(\boldsymbol{A}),\subseteq)$ for (1.) $\boldsymbol{A} = \{\boldsymbol{a}\}$ and (2.) $\boldsymbol{A} = \{\boldsymbol{a},\boldsymbol{b}\}$.

Question 4: Draw the Hasse diagram for the partial ordering $\subseteq$ on the power set $\boldsymbol{P}(\boldsymbol{S})$ where $\boldsymbol{S} = \{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\}$.

3.3 Extremal Elements: In a partially ordered set (poset), extremal elements are the elements that occupy the extreme positions with respect to the partial order. They include: Minimal element, Maximal element, Least element and Greatest element.

Minimal and Maximal:

Let S ⟶ Any Poset.

An element a in S is called a minimal element if no other element of S strictly precedes (is less than) a . Geometrically speaking, a is a minimal element if no edge enters a (from below).

An element b in S is called a maximal element if no element of S strictly succeeds (is larger than) b. Geometrically speaking, b is a maximal element if no edge leaves b (in the upward direction).

Note:

(i). The minimal and maximal members of a partially ordered set need not unique.

(ii). Maximal and minimal elements are easily calculated from the Hasse diagram. They are the *'top'* and *'bottom'* elements in the diagram.

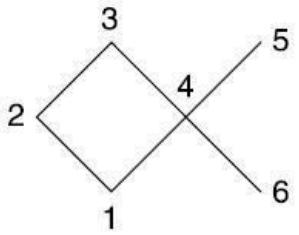
Least and Greatest Element: An element a is S is called a first (least) element if for every element x in S,

$a \preccurlyeq x$ that is, if a precedes every other element in S.

Similarly, an element b in S is called a last element if for every element y in S,

$y \preccurlyeq b$ that is, if b succeeds every other element in S.

Example1: Find the maximal and minimal element for the following Hasse diagram



Solution: In the Hasse diagram, there are two maximal elements and two minimal elements. The elements 3, 5 are maximal and the elements 1 and 6 are minimal.

Example2: Let $\boldsymbol{A} = \{\mathbf{1}, \mathbf{2}, \mathbf{3}, \mathbf{4}, \mathbf{6}, \mathbf{8}, \mathbf{9}, \mathbf{12}, \mathbf{18}, \mathbf{24}\}$ be ordered by the relation "*x* divides *y*." By the Hasse diagram find the maximal, minimal, first and last element.

Solution : For Hasse diagram see example 3 of 3.2 **.** A has two maximal elements 18 and 24 and neither is a last element. A has only one minimal element, 1, which is also a first element.

Example3: Let B ={1,2,3,4,5,6} and "/ " is a partial order relation on B . By the Hasse diagram find the maximal, minimal, first and last element.

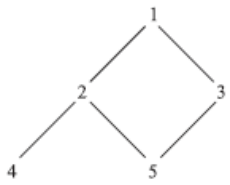
. **Solution :**For Hasse diagram see example 2 of 3.2 **.** B has two minimal elements, d and e, and neither is a first element. B has only one maximal element, a, which is also a last element .

Example 4 : Consider the set $A = \{x, y, z, u, v\}$ with the order relation $x < y < z < u < v$. By the Hasse diagram find the maximal, minimal, first and last element.

Solution : For Hasse diagram see example 4 of 3.2 **.** The chain has one minimal element, x, which is a first element, and one maximal element, v, which is a last element.

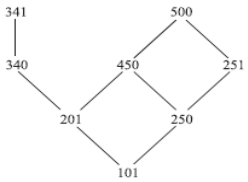
Practice Questions:

Question 1: Let $A = \{1, 2, 3, 4, 5\}$ be ordered by the Hasse diagram

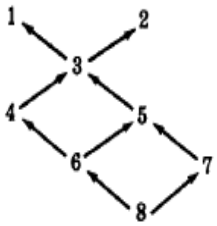


Find all minimal and maximal elements of A.
Question 2: Does A of question 1 have a first element or a last element ?

Question 3: Find the maximal, minimal, first and last element in the following Hasse diagram



Question 4: Find the maximal, minimal, first and last element in the following Hasse diagram



3.4 Supremum And Infimum:

Let S ⟶ any Poset .

A ⟶ any subset of S.

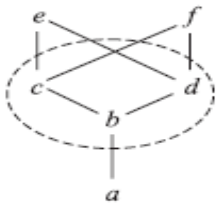
An element M in S is called an upper bound of A if M succeeds every element of A, i.e., if, for every x in A, we have

$$x \preccurlyeq M$$

If an upper bound of A precedes every other upper bound of A, then it is called the supremum of A and is denoted by sup(A)

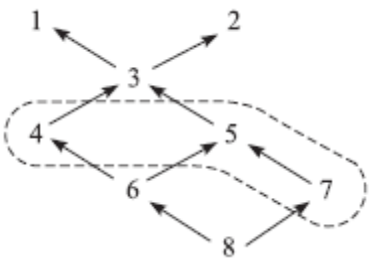
An element m in a Poset S is called a lower bound of a subset A of S if m precedes every element of A, i.e., if, for every y in A , we have $m \preccurlyeq y$
If a lower bound of A succeeds every other lower bound of A, then it is called the infimum of A and is denoted by inf (A).

Example1: Let $\boldsymbol{S} = \{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}, \boldsymbol{d}, \boldsymbol{e}, \boldsymbol{f}\}$ be ordered as pictured in Fig., and let $\boldsymbol{A} = \{\boldsymbol{b}, \boldsymbol{c}, \boldsymbol{d}\}$. Then find the upper bound ,lower bound ,supremum and infinmum of A .



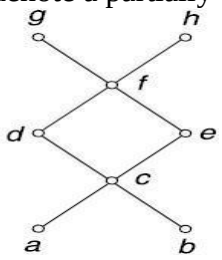
Solution: The upper bounds of A are e and f since only e and f succeed every element in A. The lower bounds of A are a and b since only a and b precede every element of A. Note that e and f are noncomparable; hence sup(A) does not exist. However, b also succeeds a, hence inf (A) = b.

Example2: Let $\boldsymbol{S} = \{\boldsymbol{1}, \boldsymbol{2}, \boldsymbol{3}, \ldots, \boldsymbol{8}\}$ be ordered as pictured in Fig. and let $\boldsymbol{A} = \{\boldsymbol{4}, \boldsymbol{5}, \boldsymbol{7}\}$. Then find the upper bound ,lower bound ,supremum and infinmum of A .



Solution: The upper bounds of A are 1, 2, and 3, and the only lower bound is 8. Note that 7 is not a lower bound since 7 does not precede 4. Here sup(A) = 3 since 3 precedes the other upper bounds 1 and 2. Note that inf (A) = 8 since 8 is the only lower bound.

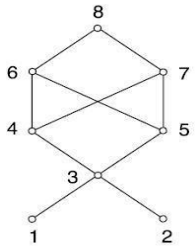
Example3: Let $\boldsymbol{A} = \{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}, \boldsymbol{d}, \boldsymbol{e}, \boldsymbol{f}, \boldsymbol{g}, \boldsymbol{h}\}$ denote a partially ordered set whose Hasse diagram is shown in



If $\boldsymbol{A} = \{\boldsymbol{c}, \boldsymbol{d}, \boldsymbol{e}\}$ then find the upper bound ,and least upper bound of A .

Solution: f, g, h are upper bounds of B. The element f is least upper bound.

Example 4: Consider the poset $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$ whose Hasse diagram is shown in Fig and

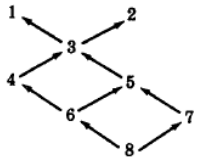


let $A = \{3, 4, 5\}$ then find the lower bounds ,and greatest lower bound of A .

Solution: The elements 1, 2, 3 are lower bounds of B. 3 is greatest lower bound.

Practice Questions:

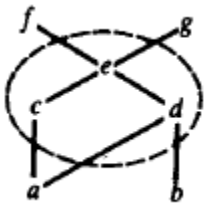
Question 1: Let $\boldsymbol{S} = \{\boldsymbol{1}, \boldsymbol{2}, \boldsymbol{3}, \ldots, \boldsymbol{8}\}$ be ordered as in Fig. , and let $\boldsymbol{A} = \{\boldsymbol{2}, \boldsymbol{3}, \boldsymbol{6}\}$



Find the upper and lower bounds of A. Identify sup(A) and inf (A) if either exists.

Question 2: Consider the set B ={1,2,5} in the above question in place $\mathbf{A} = \{\mathbf{2}, \mathbf{3}, \mathbf{6}\}$. Then find the upper bound ,lower bound ,supremum and infinmum of B .

Question 3: Let $\boldsymbol{S} = \{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}, \boldsymbol{d}, \boldsymbol{e}, \boldsymbol{f}, \boldsymbol{g}\}$ be ordered as in Fig. and let $\boldsymbol{X} = \{\boldsymbol{c}, \boldsymbol{d}, \boldsymbol{e}\}$.



find the upper bound ,lower bound ,supremum and infinmum of X.

Question 4: Consider the set $\boldsymbol{Y} = \{\boldsymbol{e}, \boldsymbol{c}\}$ in the above question in place $\boldsymbol{X} = \{\boldsymbol{c}, \boldsymbol{d}, \boldsymbol{e}\}$. Then find the upper bound ,lower bound ,supremum and infinmum of Y .

3.5 LATTICES

A lattice *L* may be defined as a partially ordered set in which $\boldsymbol{inf}\ (\boldsymbol{a}, \boldsymbol{b})$ and $\boldsymbol{sup}(\boldsymbol{a}, \boldsymbol{b})$ exist for any pair of elements $\boldsymbol{a}, \boldsymbol{b} \in \boldsymbol{L}$.

Another way is to define a lattice L axiomatically. This we do below.

Axioms Defining a Lattice

Let L be a nonempty set closed under two binary operations called meet and join, denoted respectively by ∧ and ∨ . Then L is called lattice if the following axioms hold where a, b, c are elements in L:

[1] Commutative law:

(1a) $a \wedge b = b \wedge a$ (1b) $a \vee b = b \vee a$

[2] Associative law:

(2a) $(a \wedge b) \wedge c = a \wedge (b \wedge c)$ (2b) $(a \vee b) \vee c = a \vee (b \vee c)$

[3] Absorption law:

(3a) $a \wedge (a \vee b) = a$ (3b) $a \vee (a \wedge b) = a$

We will sometimes denote the lattice by $(L,\wedge,\vee)$ when we want to show which operations are involved.

3.6 Properties of Lattice:

A lattice is a partially ordered set $(L, \leq)$ in which every pair of elements $a, b \in L$ has a Greatest Lower Bound (GLB) and a Least Upper Bound (LUB). The Greatest Lower Bound (GLB) of a subset $\{a, b\} \subseteq L$ will be denoted by $a \wedge b$ and the Least Upper Bound (LUB) by $a \vee b$. So GLB $\{a, b\} = a \wedge b$, called the meet of a and b and LUB $\{a, b\} = a \vee b$, called the join of a and b.Note that ∧ and ∨ are binary operations and we denote the lattice by (∧ ,∨).

From the definition of ∧ and ∨, the following is denoted:

(i) $a \leq a \vee b$; $b \leq a \vee b$ (i.e., $a \vee b$ is an UB of a and b).

(ii) $a \wedge b \leq a$; $a \wedge b \leq b$ (i.e., $a \wedge b$ is a LB of a and b).

(iii) If $a \leq c$ and $b \leq c$ then $a \vee b \leq c$ (i.e., $a \vee b$ is the LUB of a and b).

(iv) If $c \leq a$ and $c \leq b$ then $c \leq a \wedge b$. (i.e., $a \wedge b$ is the GLB of a and b).

Example1: Let Z^+ denote the set of all positive integers and let *R* denote the relation 'division' in Z^+, such that for any two elements *a*, $b \in Z$, *aRb*, if *a* divides *b*. Then show that (Z^+, *R*) is a lattice.

Solution : Let Z^+ denote the set of all positive integers and let *R* denote the relation 'division' in Z^+, such that for any two elements *a*, $b \in Z$, *aRb*, if *a* divides *b*. Then (Z^+, *R*) is a lattice in which the join of *a* and *b* is the least common multiple of *a* and *b*, i.e. $a \vee b = a \oplus b$ = LCM of *a* and *b*, Since the LCM of any two positive integers always exists and is unique, every pair has a **Join** and the meet of *a* and *b*, i.e. $a * b$ is the greatest common divisor (GCD) of *a* and *b* i.e., $a \wedge b = a * b$ = GCD of *a* and *b*. Since the GCD of any two positive integers always exists and is unique, every pair has a Meet.

We can also write $\boldsymbol{a + b = a \vee b = a \oplus b}$ =LCM of *a* and *b* and $\boldsymbol{a.b = a \wedge b = a * b}$ =GCD of *a* and *b*.

Question2: Let A={1,2,3,6} be a partially ordered set under the divisibility relation (/). Show that (A,/) is a Lattice.

Solution: A partially ordered set (A, ≤) is called a Lattice if every pair of elements has a Greatest Lower Bound (GLB) or Meet (∧)and a Least Upper Bound (LUB) or Join (∨). That is,
Every pair of elements must have both Meet and Join.

For the divisibility relation,

Meet = HCF (Greatest Common Divisor)

Join = LCM (Least Common Multiple)

Pair	Meet (HCF)	Join (LCM)
(1,2)	1	2
(1,3)	1	3
(1,6)	1	6
(2,3)	1	6
(2,6)	2	6
(3,6)	3	6

Since both Meet and Join exist for every pair of elements, the condition of a lattice is satisfied.

Every pair of elements in the given POSET has a Greatest Lower Bound (Meet) and a Least Upper Bound (Join**)**. Hence, (A, /) is a Lattice.

Question3: Let A={∅,{a},{b},{a,b}} be partially ordered by the subset relation (⊆).Show that (A,⊆) is a Lattice .

Solution: A partially ordered set (A,≤) is called a **Lattice** if every pair of elements has a Greatest Lower Bound (Meet), and a Least Upper Bound (Join).

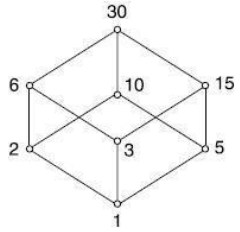
For the subset relation, Meet = Intersection (∩) and Join = Union (∪)

Pair of Elements	**Meet (∩)**	**Join (∪)**
∅,{a}	∅	{a}
∅,{b}	∅	{b}
∅,{a,b}	∅	{a,b}
{a},{b}	∅	{a,b}
{a},{a,b}	{a}	{a,b}
{b},{a,b}	{b}	{a,b}

From the above table, every pair of elements has a unique Meet**,** and a unique Join.

Hence, the definition of a lattice is satisfied. Therefore, (A,⊆) is a Lattice.

Example4: Let n be a positive integer and let S_n be the set of all positive divisors of n. If n=30, , Then S_{30}={1,2,3,5,6,10,15,30}. Let R be the divisibility relation defined by aRb ⟺ a|b. Show that (S_{30},R) is a lattice.



Solution: A partially ordered set (P, ≤) is called a lattice if every pair of elements has a Greatest Lower Bound (Meet) and a least Upper Bound (Join). In the divisibility relation, Meet = HCF (GCD) and Join = LCM

Consider some pairs of elements of S_{30}.

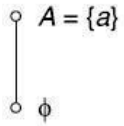
Pair	Meet (HCF)	Join (LCM)
(2,3)	1	6
(2,5)	1	10
(3,5)	1	15
(6,10)	2	30
(6,15)	3	30
(10,15)	5	30

From the above table, the Meet (HCF) and Join (LCM) of every pair of elements belong to S_{30}.
Hence, every pair of elements has both a Meet and a Join.
Therefore, (S_{30}.,R) is a Lattice.

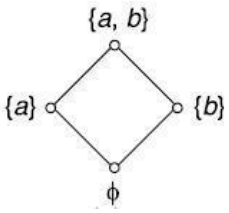
Example 5: Let *A* be any set and *P* (*A*) be its power set. The poset (*P* (*A*), ⊆) is a lattice in which the

meet and join are the same as the operations ∩ and ∪ on sets respectively.

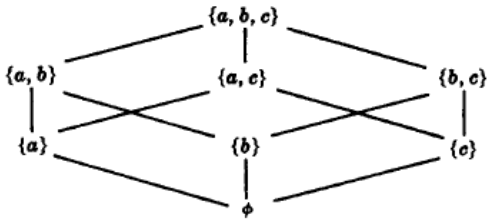
$S = \{a\}$, $P(A) = \{\phi, \{a\}\}$



$S = \{a, b\}$, $P(A) = \{\phi, \{a\}, \{b\}, S\}$.



Example 6: Let C be a collection of sets closed under intersection and union. Then $(C,\cap,\cup)$ is a lattice. In this lattice, the partial order relation is the same as the set inclusion relation. Following figure shows the diagram of the lattice L of all subsets of $\{a,b,c\}$.



Practice Question :

Question 1: Prove that (S,≤) is a Lattice, where S= {1,2,3,6} and ≤ is for divisibility .

Question 2 : Let A={1,2,4,8} , define the relation R on A by aRb ⟺ a /b (where a divides b). Show that (A,R) is a Lattice.

Question 3: Let P(S)={ ϕ,{a},{b},{a,b}} where S={a,b} and the partial order is set inclusion (⊆) Show that (P(S),⊆)(P(S), is a Lattice.

Question 4: Let A={1,2,3} with the relation R={(1,1),(2,2),(3,3),(1,2),(1,3)}. Determine whether (A,R) is a Lattice.

Theorems 1: Let $(L,\leq)$ be a lattice in which ∧ and ∨ denote the operations of meet and join respectively. For any $a,b,c \in L$, we have
(i) $a \wedge a = a$; $a \vee a = a$
(Idempotent)
(ii) $a \wedge b = b \wedge a$; $a \vee b = b \vee a$
(Commutative)
(iii) $(a \wedge b) \wedge c = a \wedge (b \wedge c)$; $(a \vee b) \vee c = a \vee (b \vee c)$
(Associative)
(iv) $a \wedge (a \vee b) = a$; $a \vee (a \wedge b) = a$
(Absorption)

Proof: Following are the proof of above mentioned statements:

(i) Since a ≤ a, we know that a is a lower bound of {a, a} = {a}. If b is also a lower bound and if a ≤ b then we have b ≤ a and a ≤ b. By antisymmetry, a = b. So a is the GLB of {a}. Therefore, a ∧ a = a. Dually, a ∨ a = a follows.

(ii) Let $x = a \wedge b = GLB\{a,b\}$. Since $\{a,b\} = \{b,a\}$, GLB $\{b,a\} = x$. So

$b \wedge a = x$. Hence $x = a \wedge b = b \wedge a$. Dually, $a \vee b = b \vee a$ follows.

(iii)Let $x = a \wedge (b \wedge c)$ and $y = (a \wedge b) \wedge c$.
Now $x = a \wedge (b \wedge c) \Longrightarrow x \leq a, x \leq b \wedge c$.
$\Longrightarrow x \leq a, x \leq b, x \leq c$
$\Longrightarrow x \leq a \wedge b, x \leq c$
$\Longrightarrow x \leq (a \wedge b) \wedge c = y$.
Similarly, $y \leq x$ follows. By antisymmetry, x = y and hence,
$a \wedge (b \wedge c) = (a \wedge b) \wedge c$.
Dually, $a \vee (b \vee c) = (a \vee b) \vee c$ follows.

(iv)By definition, for any $a \in L, a \leq a \; and \; a \leq a \vee b$
Therefore, $a \leq a \wedge (a \vee b)$. But $a \wedge (a \vee b) \leq a$. Hence $a \wedge (a \vee b) = a$.
Dually, $a \vee (a \wedge b) = a$ follows.

Theorem2: Let (L, ≤) be a lattice in which ∧ and ∨ denote the operations of meet and join, respectively. For any $a, b \in L$,
$a \leq b \Leftrightarrow a \wedge b = a$
$\Leftrightarrow a \vee b = b$
Proof:Assume that $a \leq a. Since \; a \leq b$, it follows that $a \leq a \wedge b$. But by definition
of ∧ , a ∧ b ≤ a. Therefore $a \wedge b = a$. Conversely, suppose $a \wedge b = a$. Then a ≤
b. Hence $a \leq b \Leftrightarrow a \wedge b = a$. Similarly $a \leq b \Leftrightarrow a \vee b = b$ follows.

Isotonicity law

Theorem3: Let (L, ≤) be a lattice in which ∧ and ∨ denote the operations of meet and join, respectively. For any a, b, c, ∈ L,
$$b \leq c \Longrightarrow \begin{cases} a \wedge b \leq a \wedge c \\ a \vee b \leq a \vee c \end{cases}$$

Proof: Assume that b ≤ c. Since a ∧b ≤ b, by transitivity, $a \wedge b \leq c$. Since $a \wedge b$
$\leq a$, it follows that, $a \wedge b \leq a \wedge c$
Now, $b \leq c$ and $c \leq a \vee c$ implies $b \leq a \vee c$. But $a \leq a \vee c$. Hence $a \vee b \leq a \vee c$.
Note: For any $a, b, c \in L$, by Isotonicity law,
$a \leq b \wedge a \leq c \Longrightarrow a \leq b \vee c$
$a \leq b \wedge a \leq c \Longrightarrow a \leq b \vee c$

$$c \leq b \wedge a \leq a \Longrightarrow b \wedge c \leq a$$
$$c \leq b \wedge a \leq a \Longrightarrow b \vee c \leq a.$$

Distributive Inequality

Theorem4: Let $(L, \leq)$ be a lattice. For any $a, b, c, \in L$, the following inequalities are hold.
(i) $a \vee (b \wedge c) \leq (a \vee b) \wedge (a \vee c)$
(ii) $(a \wedge b) \vee (a \wedge c) \leq a \wedge (b \vee c)$
Since $a \leq a \vee b$ *and* $a \leq a \vee c$, we have,
$a \leq (a \vee b) \wedge (a \vee c) \ldots (1)$
Since $b \wedge c \leq b \leq a \vee b$ *and* $b \wedge c \leq c \leq a \vee c$,
$b \wedge c \leq (a \vee b) \wedge (a \vee c) \ldots (2)$
From Equations (1) and (2) we have,
$a \vee (b \wedge c) \leq (a \vee b) \wedge (a \vee c)$.
Similarly, Case (ii) follows.

Modular Inequality

Theorem5: Let $(L, \leq)$ be a lattice. For any $a, b, c \in L$,
$a \leq c \Leftrightarrow a \vee (b \wedge c) \leq (a \vee b) \wedge c$
Proof: Suppose $a \leq c$. Then $a \vee c = c$.
By distributive inequality, $a \vee (b \wedge c) \leq (a \vee b) \wedge (a \vee c)$
Since $a \vee c = c$,
$a \vee (b \vee c) \leq (a \vee b) \vee c$.
Conversely, let us assume that $a \vee (b \wedge c) \leq (a \vee b) \wedge c$.
Since,
$$a \leq a \vee (b \wedge c)$$
$$\leq (a \vee b) \wedge c$$
$$\leq c$$
We get a ≤ c. Hence proved.

Theorem6: Prove that in a lattice $(L, \leq)$, for any $a, b, c \in L$, if $a \leq b \leq c \Longrightarrow a \vee b = bc$, and $(a \wedge b) \vee (b \wedge c) = b = (a \vee b) \wedge (a \vee c)$.
Solution: Since $a \leq b$ and $a \leq c, a \leq b \wedge c$. Again $b \leq b$ and $b \leq c$ implies $b \leq b \wedge c$.
Now $a \leq b \wedge c$ *and* $b \leq b \wedge c \Longrightarrow a \vee b \leq b \wedge c$ ... (1)
Again, $b \wedge c \leq b \leq a \vee b$ ... (2)
From Equations (1) and (2), $a \vee b = b \wedge c$.
Hence, $a \wedge b \leq b$ *and* $b \wedge c \leq b$, *we get* $(a \wedge b) \vee (b \wedge c) \leq b$.
Since $b \leq c$ and
$b \leq b$ *implies* $b \leq b \wedge c$. Again this implies $b \leq (b \wedge c) \vee (a \wedge b)$. Hence, $(a \wedge b) \vee (b \wedge c) = b$. Similarly, $(a \vee b) \wedge (a \vee c) = b$ follows.

Remark: Let P be a poset such that the inf (a, b) and sup(a, b) exist for any a, b in P. Letting

$a \wedge b = inf\ (a, b)$ and $a \vee b = sup(a, b)$ we have that $(P, \wedge, \vee)$ is a lattice. Furthermore, the partial order on P induced by the lattice is the same as the original partial order on P.

The converse of the above is also true. That is, let L be a lattice and let $\preccurlyeq$ be the induced partial order on L. Then $inf\ (a, b)\ and\ sup(a, b)$ exist for any pair $a, b\ in\ L$ and the lattice obtained from the poset (L, $\preccurlyeq$) is the original lattice. Accordingly, we have the following:

Alternate Definition: A lattice is a partially ordered set in which $a \wedge b = inf\ (a, b)$ and $a \vee b = sup(a, b)$ exist for any pair of elements a and b. We note first that any linearly ordered set is a lattice since $inf\ (a, b) = a$ and $sup(a, b) = b$ whenever a $\preccurlyeq$b.

3.7 BOUNDED LATTICES

A lattice *L* is said to have a lower bound 0 if for any element x in *L* we have 0 $\preccurlyeq$ x. Analogously, L is said to have an upper bound I if for any x in *L* we have x $\preccurlyeq$ I . We say *L* is bounded if *L* has both a lower bound 0 and an upper bound I . In such a lattice we have the identities $a \vee I = I, a \wedge I = a, a \vee 0 = a, a \wedge 0 = 0$ for any element a in L.

The nonnegative integers with the usual ordering, $0 < 1 < 2 < 3 < 4 <$ ... ... have 0 as a lower bound but have no upper bound. On the other hand, the lattice P(U) of all subsets of any universal set U is a bounded lattice with U as an upper bound and the empty set ϕ as a lower bound. Suppose $L = \{a_1, a_2, \ldots, a_n\}$ is a finite lattice. Then $a_1 \vee a_2 \vee$ ... ... $\vee a_n\ and\ a_1 \wedge a_2 \wedge$ ... .$\wedge a_n$ are upper and lower bounds for *L*, respectively.

Remark : Every finite lattice L is bounded.

3.8 DISTRIBUTIVE LATTICES

A lattice *L* is said to be distributive if for any elements a, b, c in *L* we have the following:

Distributive law:

$$(i)\ a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)\ (ii)\ a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

Otherwise, *L* is said to be non distributive. We note that by the principle of duality the condition (i) holds if and only if (ii) holds.

Figure 3.8(a) is a nondistributive lattice since

$a \vee (b \wedge c) = a \vee 0 = a \, but \, (a \vee b) \wedge (a \vee c) = I \wedge c = c$

Figure 3.8 (b) is also a non distributive lattice.

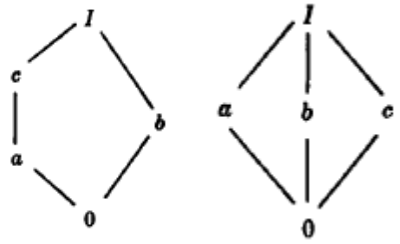


Fig .3.8 (a) **Fig .3.8 (b)**

3.9 COMPLEMENTS , COMPLEMENTED LATTICES

Let L be a bounded lattice with lower bound 0 and upper bound I . Let a be an element of L. An element x in L is called a complement of a if $a \vee x = I$ and $a \wedge x = 0$.

Complements need not exist and need not be unique. For example, the elements a and c are both complements of b in Fig. 3.9 (a). Also, the elements y, z, and u in the chain in Fig.3.9 (b) have no complements. We have the following result..

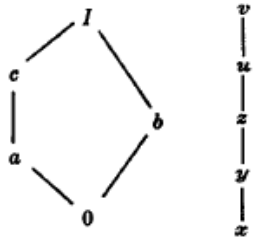


Fig 3.9 (a) Fig 3.9 (b)

Theorem 1: Let L be a bounded distributive lattice. Then complements are unique if they exist.

Proof: Suppose x and y are complements of any element a in L. Then

$$a \vee x = I, a \vee y = I, a \wedge x = 0, a \wedge y = 0$$

Using distributivity,

$$x = x \vee 0 = x \vee (a \wedge y) = (x \vee a) \wedge (x \vee y) = I \wedge (x \vee y) = x \vee y$$

Similarly,

$$y = y \vee 0 = y \vee (a \wedge x) = (y \vee a) \wedge (y \vee x) = I \wedge (y \vee x) = y \vee x$$

Thus

$$x = x \vee y = y \vee x = y$$

and the theorem is proved.

Complemented Lattices

A lattice *L* is said to be complemented if *L* is bounded and every element in *L* has a complement. Figure 3.8(b) shows a complemented lattice where complements are not unique. On the other hand, the lattice *P(U)* of all subsets of a universal set U is complemented, and each subset A of U has the unique complement $A^c = U \setminus A$.

3.10 Modular Lattice:

A Lattice $(L, \leq)$ is said to be modular lattice if

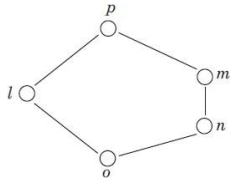
$$a \leq c \Rightarrow a \vee (b \wedge c) = (a \vee b) \wedge c.$$

Example1: Show that the following simple but significant lattices are not distributive.



Solution :(a) To see that the diamond lattice is not distributive, use the middle elements of the
lattice: $a \wedge (b \vee c) = a \wedge 1 = a, but\ (a \wedge b) \vee (a \wedge c) = 0 \vee 0 = 0, and\ a \neq 0.$
Similarly, the other distributive law fails for these three elements.(b) The pentagon lattice is also not distributive

Example2: Show that lattice is not a distributive lattice.



Solution :A lattice is distributive if all of its elements follow distributive property so let we verify the distributive property between the elements *n*, *l* and *m*.

$$GLB(n, LUB(l,m)) = GLB(n,p)\ [\because LUB(l,m) = p]$$
$$= n\ (LHS)$$

also $LUB(GLB(n,l), GLB(n,m)) = LUB(o,n);\ [\because GLB(n,l) = o\ and\ GLB(n,m) = n]$

$$= n\ (RHS)$$

so LHS = RHS.

But $GLB(m, LUB(l,n)) = GLB(m,p)\ [\because LUB(l,n) = p]$

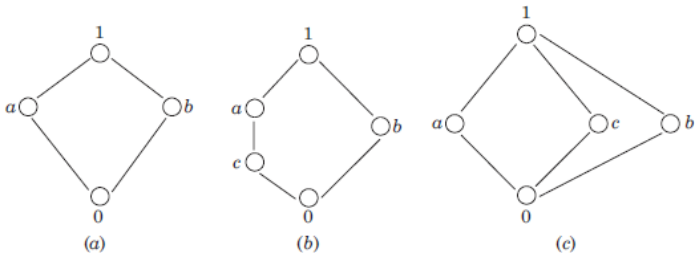
$$= m\ (LHS)$$

also $LUB(GLB(m,l), GLB(m,n)) = LUB(o,n);\ [\because GLB(m,l) = o\ and\ GLB(m,n) = n]$

$= n\ (RHS)$

Thus, LHS ≠ RHS hence distributive property doesn't hold by the lattice so lattice is not distributive.

Example3: Lattices shown in Fig (a), (b) and (c) are complemented lattices.



Solution : For the lattice (*a*) $GLB(a,b) = 0\ and\ LUB(x,y) = 1$. So, the complement *a* is *b* and vise versa. Hence, a complement lattice.
For the lattice (*b*) $GLB(a,b) = 0\ and\ GLB(c,b) = 0\ and\ LUB(a,b) = 1\ and\ LUB(c,b) = 1$; so both *a* and *c* are complement of *b*. Hence, a complement lattice.
In the lattice (*c*) $GLB(a,c) = 0\ and\ LUB(a,c) = 1;\ GLB(a,b) = 0\ and\ LUB(a,b) = 1$. So,
complement of *a* are *b* and *c*.
Similarly complement of *c* are *a* and *b* also *a* and *c* are complement of *b*. Hence lattice is a complement lattice.

Example4: Consider the poset (X, ≤) where X = {1, 2, 3, 5, 30} and the partial ordered relation ≤is defined as i.e. if x and y ∈X then x ≤ y means ‗x divides y'. Then show that poset (I+, ≤) is a lattice but not a distributive lattice .
Solution : Since GLB(*x*, *y*) = *x* ∧ *y* = *lcm*(*x*, *y*) and LUB(*x*, *y*) = *x* ∨ *y* = *gcd*(*x*, *y*)
Now we can construct the operation table I and table II for GLB and LUB respectively and the

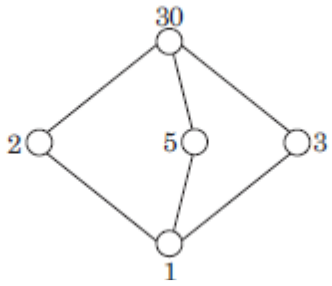
Hasse diagram is shown in Fig

Table I

LUB	1	2	3	5	30
1	1	2	3	5	30
2	2	2	30	30	30
3	3	30	3	30	30
5	5	30	30	5	30
30	30	30	30	30	30

Table II

GLB	1	2	3	5	3
1	1	1	1	1	
2	1	2	1	1	
3	1	1	3	1	
5	1	1	1	5	
30	1	2	3	5	3



Test for distributive lattice, i.e.,

$$GLB(x, LUB(y,z)) = LUB(GLB(x,y), GLB(x,z))$$

Assume $x = 2, y = 3$ *and* $z = 5$, then

RHS: $GLB(2, LUB(3,5)) = GLB(2,30) = 2$

LHS: $LUB(GLB(2,3), GLB(2,5)) = LUB(1,1) = 1$

Since *RHS* $\neq$ *LHS*, hence lattice is not a distributive lattice.

Practice Question :

Question 1: Prove that $(S, \leq)$ is a Lattice, where S= {1,2,5,10} and $\leq$ is for divisibility. Prove that it is also a Distributive Lattice?

Question 2: Prove that $(S, \leq)$ is a Lattice, where $S = \{1,2,3,6\}$ and $\leq$ is for divisibility. Prove that it is also a Distributive Lattice?

Question 3: Show that in a lattice with two or more elements, no element is its own complement.

Question 4: A lattice is said to the modular if $a \leq c \Rightarrow a \vee (b \wedge c) = (a \vee b) \wedge c$. Showthat every distributive lattice is modular but not conversely.

Question 5: Show that a lattice is modular if and only if $a \vee (b \wedge (a \vee c)) = (a \vee b) \wedge (a \vee c)$.

Is the cartesian product of two lattices always a lattice? Prove your claim.

3.11.Boolean Algebras and Applications:

Boolean Algebra: A Boolean algebra is a set with two special elements 0 and 1; two binary operations denoted by + (sum) and · (product); and, a unary operation denoted by $'$ (complementation) such that, for all $a, b, c \in B$, the following five axioms hold:

B1. (Identity Laws) $a + 0 = a \; and \; a \cdot 1 = a$;
B2. (Commutative Laws) $a + b = b + a \; and \; a \cdot b = b \cdot a$;
B3. (Associative Laws) a+$(b + c) = (a + b) + c \; and \; a \cdot (b \cdot c) = (a \cdot b) \cdot c$;
B4. (Distributive Laws) $a \cdot (b + c) = a \cdot b + a \cdot c \; and \; a + (b \cdot c) = (a + b) \cdot (a + c)$;
B5. (Complement Laws) $a + a' = 1 \; and \; a \cdot a' = 0$.
We write a Boolean algebra as$(B, +, \cdot, ', 0, 1)$, where 0 is called the zero element and 1 is called the identity element. Also, for $a \in B$, the element $a' \in B$ is called the complement of a.

Important Boolean Algebras:

Example 1: The algebra of sets P(X) defined on a nonempty set X is a Boolean algebra, with respect to binary operations ∪and ∩; usual "**set complementation**" as unary operation; and, by taking 0 ≡ ∅, 1 ≡ X . This is known as **the Boolean algebra of sets** given by the set X.

Example2: Lattice Boolean Algebra: Let L, $\leq$, ∨, ∧ be a complemented distributive lattice, with bounds denoted by 0 and 1, and $a, b \in L$. Since the join and meet of every pair of elements in L are unique, it follows that ∨ and ∧ are binary operations on the set L. Since L is complemented distributive lattice, the mapping $a' \rightarrow a' : L \rightarrow L$ is well defined, where a' denotes the complement of the element a. Now, taking ∨ as the sum + and ∧ as the product ·, it follows from the problems solved in the previous section that all the five axioms $(B1) - (B5)$ hold, with the least element 0 of L as the zero element, and the greatest element 1 of L as the identity element. Therefore, every complemented distributive lattice is a Boolean algebra, which is called the lattice Boolean algebra. For emphasis, we may this Boolean algebra as B_L.

Example3: Show that a lattice having **3 elements** or **5 elements cannot** be a Boolean algebra.

Solution: Recall that every Boolean algebra is a complemented distributive lattice.

Also, every Boolean algebra must contain

- a least element 0,
- a greatest element 1, and
- every element must have a unique complement.

Case 1: A 3 − *Element Lattice*: Suppose $L = \{0, x, 1\}$

Where 0 is the least element, 1 is the greatest element and x is the third element.

The Hasse diagram is

1

|

x

|

0

The complement of 0 is $0' = 1$.

The complement of 1 is $1' = 0$.

The complement x' must satisfy $x \vee x' = 1$ and $x \wedge x' = 0$.

Now check every possibility.

If $x' = 0$, then $x \vee 0 = x \neq 1$. Not possible.

If $x' = x$, then $x \vee x = x \neq 1$. Again impossible.

Hence, x has no complement.

Therefore, the lattice is **not complemented**.

Hence, A 3-element lattice cannot be a Boolean Algebra.

Similarly 5-element lattice cannot be a Boolean Algebra.

Result: A finite Boolean algebra always contains 2^n elements, where $n = 0,1,2,3, \ldots$

Thus, the possible numbers of elements are $1,\ 2,\ 4,\ 8,\ 16,\ 32, \ldots$

Since $3 \neq 2^n$ and $5 \neq 2^n$. A Boolean algebra **cannot** have 3 or 5 elements.

Example 4: Show that the set $B = \{1,2,3,6\}$ is a Boolean algebra with respect to operations as given below: $a + b = lcm\{a, b\},\ a \cdot b = hHcf\{a, b\},\ a' = \frac{6}{a}$. Further, show that the set $B = \{1, 2, 4, 8\}$ is a not a Boolean algebra with respect to $+$ and $\cdot$ as above, and $x' = \frac{8}{x}$.

Solution: we first show that the set $B = \{1,2,3,6\}$ is a Boolean algebra with respect to operations as given below: $a + b = lcm\{a, b\},\ a \cdot b = hHcf\{a, b\},\ a' = \frac{6}{a}$.

The three operations $+,\ \cdot, and\ '$, on the set B = {1, 2, 3, 6} are as given in Table.

Addition table (+ = LCM)

+	1	2	3	6
1	1	2	3	6
2	2	2	6	6
3	3	6	3	6
6	6	6	6	6

Multiplication table (· = HCF)

·	1	2	3	6
1	1	1	1	1
2	1	2	1	2
3	1	1	3	3
6	1	2	3	6

Complement table

a	a′	a + a′	a · a′
1	6	6	1
2	3	6	1
3	2	6	1
6	1	6	1

Closure: LCM and HCF of any two elements of B remain in B.

Commutative and associative laws: LCM and HCF are commutative and associative.

Identity elements: $a + 1 = a$ for HCF? Here, in Boolean notation, 1 acts as multiplicative identity because $a \cdot 6$? Actually with HCF/LCM: $a \cdot 6 = a$ is not always true, but $a \cdot 6 = a$ holds for $a \in \{1,2,3,6\}$. Also $a + 1 = a$ for LCM is false; the additive identity is 1 only in the Boolean-algebra interpretation where "+" means join. Here the least element is 1 and the greatest element is 6.

Complement law: For every $a \in B, a + a' = 6 \; and \; a \cdot a' = 1$.

Distributive law: LCM and HCF satisfy distributive laws on divisors of 6.

Hence, The set $B = \{1,2,3,6\}$ with the given operations forms a Boolean algebra.

Example 5: Let $B, +, \cdot, ', 0,1$ be a Boolean algebra, and $a, b \in B$. Show that

i) $a + 1 = 1 \; and \; a \cdot 0 = 0$.

(ii) $a \cdot a + b = a \; and \; a + a \cdot b = a$.

(iii) $a'' = a$.

(iv) $0' = 1 \; and \; 1' = 0$.

(v) $a + b = b \Leftrightarrow (a \cdot b = a \Leftrightarrow a \cdot b' = 0 \Leftrightarrow a' + b = 1$.

(vi) $ab' + a'b = 0 \Leftrightarrow a = b$.

Solution: (i) Show that $a + 1 = and \; a \cdot 0 = 0$

Proof: we have

$a + 1 = a + (a + a') = (a + a) + a' = a + a' = 1$. , Since $a + a' = 1$,

Hence, $a + 1 = 1$.

Now, $a \cdot 0 = a \cdot (a \cdot a') = (a \cdot a) \cdot a' = a \cdot a' = 0.$

Therefore, $a \cdot 0 = 0.$

(ii) Show that $a \cdot (a + b) = a \; and \; a + a \cdot b = a.$

Proof

Using the distributive law, $a \cdot (a + b) = a \cdot a + a \cdot b = a + a \cdot b = a.$

Hence, $a \cdot (a + b) = a.$

Now, $a + a \cdot b = (a + a)(a + b) = a(a + b) = a.$

Therefore, $a + a \cdot b = a.$

These are called the **Absorption Laws**.

(iii) Show that (a′)′=a.

Proof: We know $\boldsymbol{a + a' = 1}$ and $\boldsymbol{a \cdot a' = 0}$.

Thus, a satisfies the definition of the complement of a'.

Since complements are unique, $(a')' = a.$

(iv) Show that 0′=1 and 1′=0.

Proof: Since $\mathbf{0 + 1 = 1}$ and $\mathbf{0 \cdot 1 = 0,}$

the element 1 satisfies the definition of the complement of 0.

Hence, $0' = 1.$

Similarly, $1 + 0 = 1 \; and \; 1 \cdot 0 = 0.$

Therefore, $1' = 0.$

(v) Show that

$$a + b = b \Leftrightarrow a \cdot b = a \Leftrightarrow a \cdot b' = 0 \Leftrightarrow a' + b = 1.$$

$(a) \; a + b = b \Rightarrow a \cdot b = a$

Multiply both sides by a:

$$a(a + b) = ab$$
$$a = ab.$$

Hence, $ab = a.$

(b) $ab = a \Rightarrow ab' = 0$

$$ab' = (ab)b'$$
$$= a(bb')$$
$$= a \cdot 0 = 0.$$

Hence, $ab' = 0.$

(c) $ab' = 0 \Rightarrow a' + b = 1$

By De Morgan's law, $(ab')' = a' + (b')' = a' + b.$

Since $ab' = 0,$

taking complements $a' + b = 1.$

(d) $a' + b = 1 \Rightarrow a + b = b$

Multiply both sides by b:

$$b(a' + b) = b$$
$$ba' + b = b.$$

Equivalently, using the result a′+b=1, we have

$$(a' + b)' = 0$$
$$ab' = 0$$

which implies $ab = a.$

Then,

$$a + b = (ab) + b = b(a + 1) = b.$$

Hence, $a + b = b.$

Therefore, $a + b = b \iff ab = a \iff ab' = 0 \iff a' + b = 1.$

(vi) Show that

$$ab' + a'b = 0 \Leftrightarrow a = b.$$

Proof: Suppose

$$ab' + a'b = 0.$$

Since the sum is zero, each term must be zero: $ab' = 0$ and $a'b = 0.$

From part (v), $ab' = 0 \Rightarrow a + b = b,$

So $a \leq b$.

Also, $a'b = 0 \Rightarrow a + b = a,$

So $b \leq a.$

Therefore, $a = b.$

Conversely, if $a = b,$

Then $ab' + a'b = aa' + a'a = 0 + 0 = 0.$

Hence, $ab' + a'b = 0 \iff a = b.$